

What is Scrum:

Scrum meaning - Scrum is a **framework** for **developing, delivering, and sustaining** complex products. It is essentially a framework wherein people can address complex, adaptive problems. At the same time, they can deliver products of the highest possible value in a productive and creative manner.

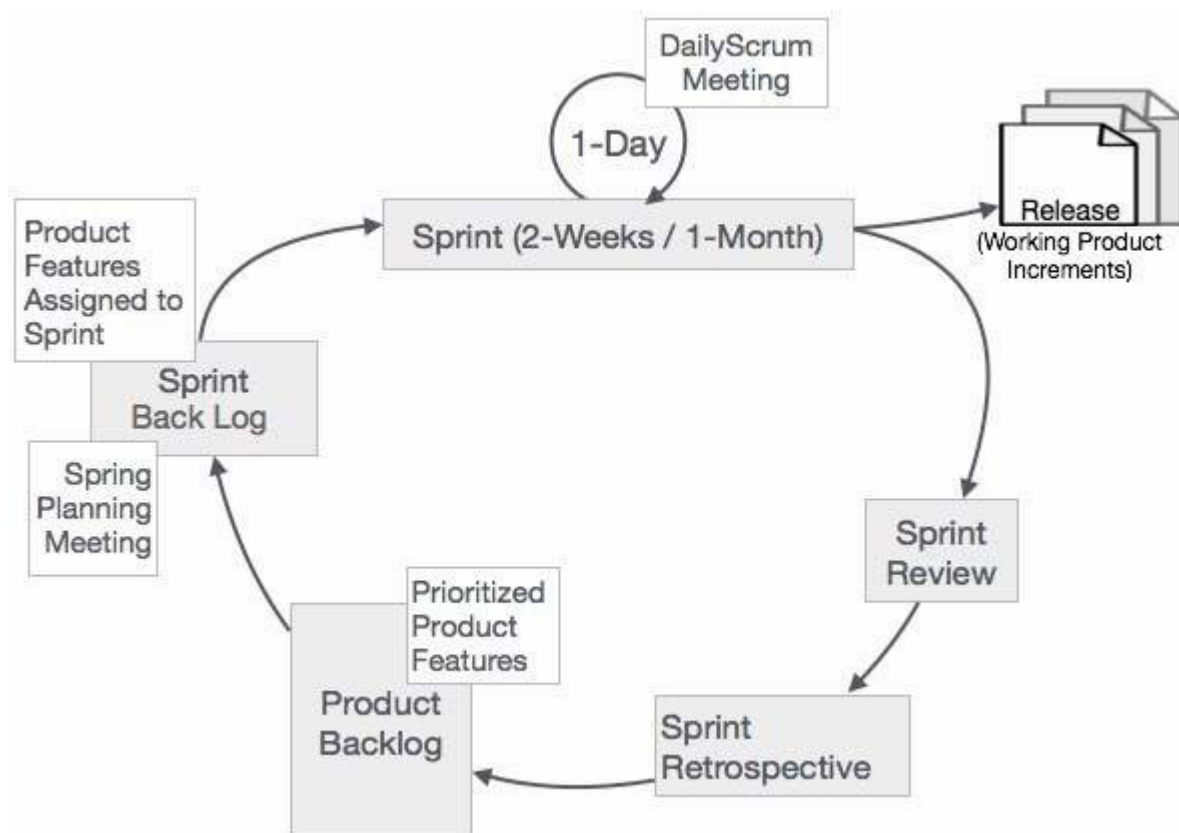
What is scrum?

Scrum is an agile project management framework that helps teams' structure and manage their work through a set of values, principles, and practices. Much like a rugby team (**where it gets its name**) training for the big game, scrum encourages teams to learn through experiences, self-organize while working on a problem, and reflect on their wins and losses to continuously improve.

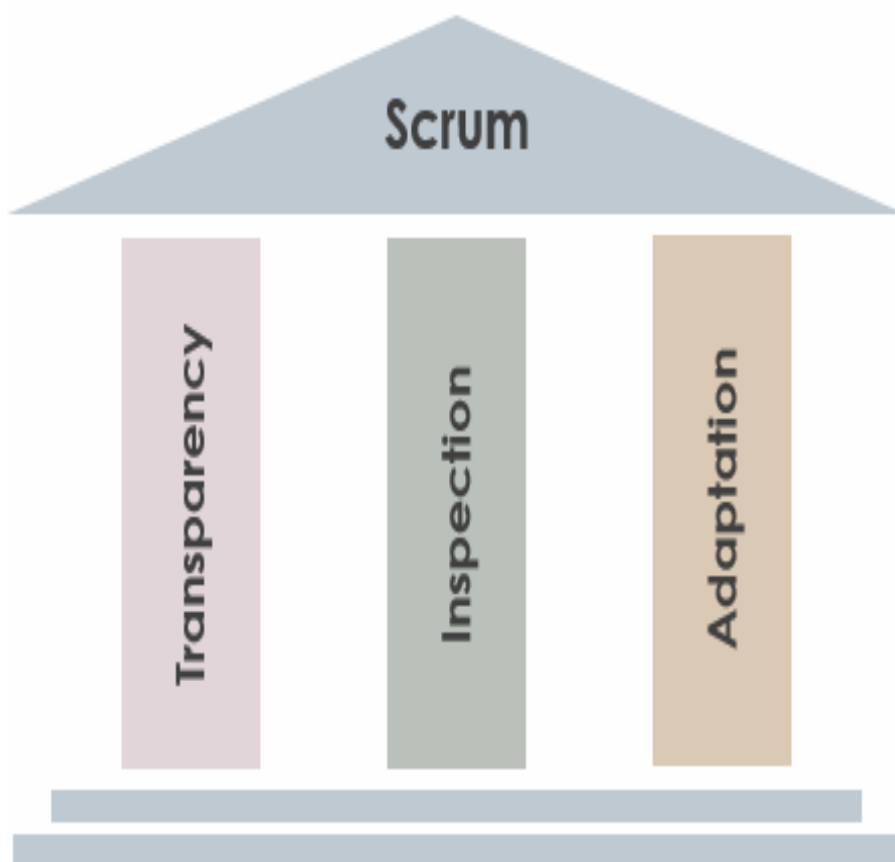
Define Scrum - What does Scrum mean?

Scrum is an iterative and incremental structure for project management mainly used in agile software development. The scrum methodology indicates ;, the **versatility** to change accompanying with emerging communication, collaboration, and business realities.

Below is the Scrum diagram or Scrum framework:



In Scrum, the prescribed events are used to create regularity. All events are time-boxed events, such that every event has a maximum duration.



Transparency

Giving visibility to the significant aspects of the process to those responsible for the outcome.

Inspection

Timely checks on the progress toward a sprint goal to detect undesirable variances.

Adaptation

Adjusting a process as soon as possible to minimize any further deviation or issues.

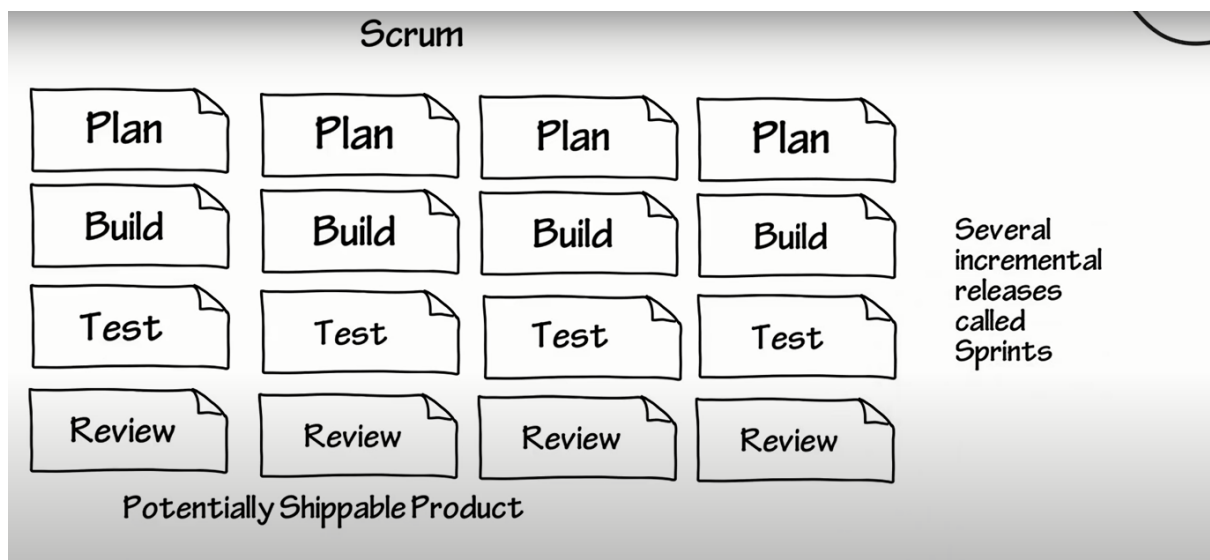
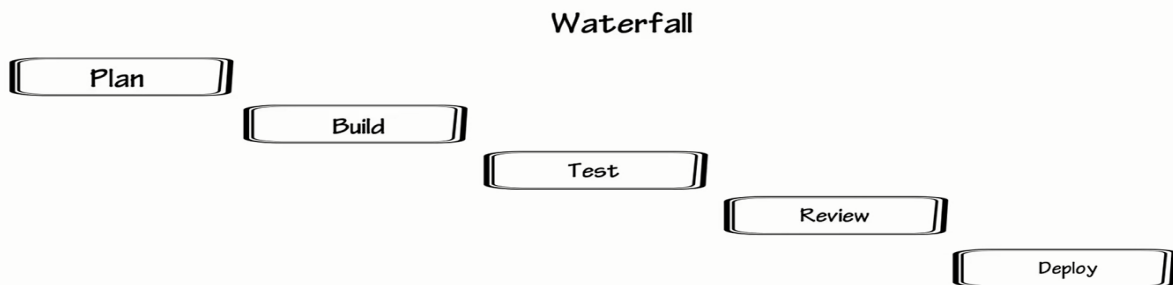
Uses of Scrum :

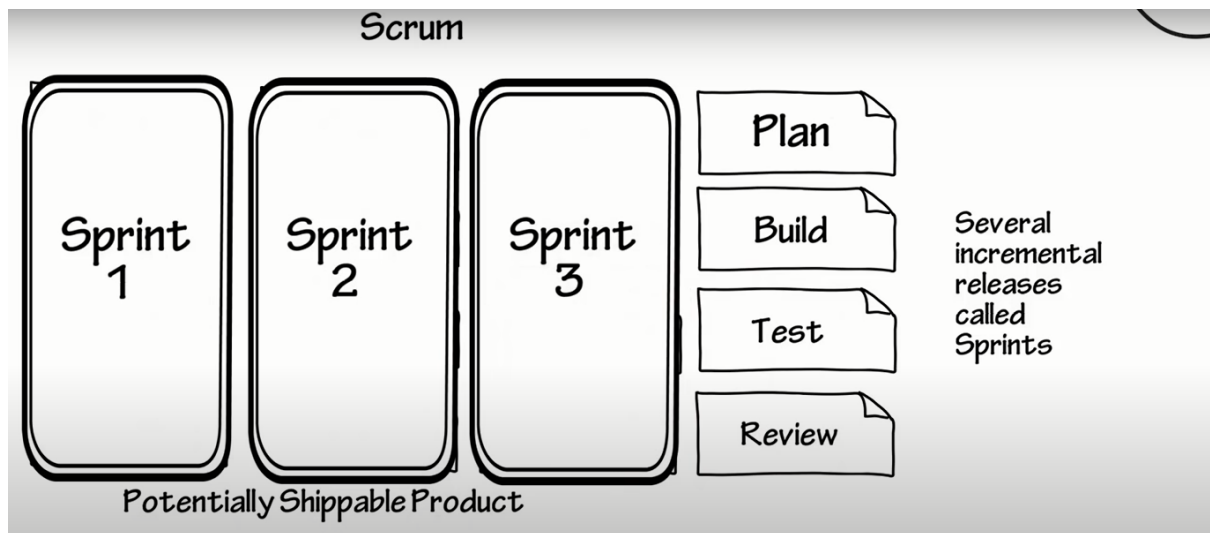
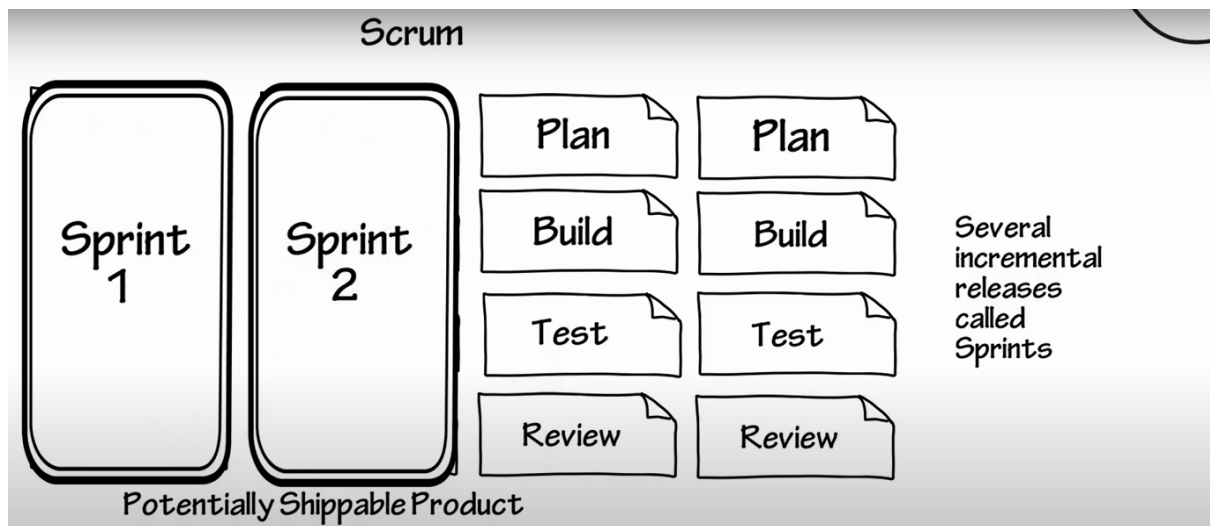
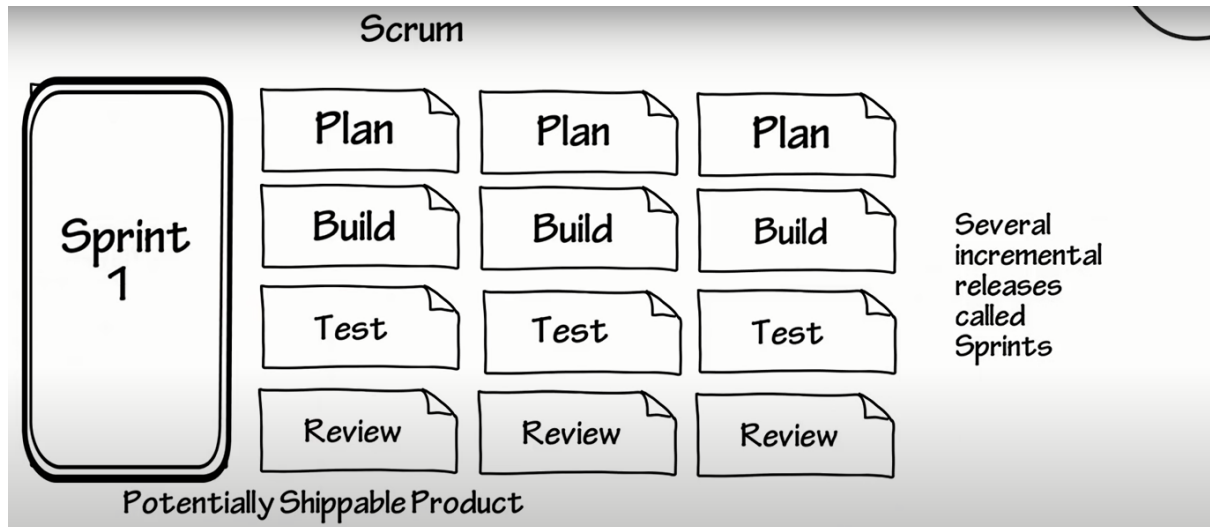
An organization implementing Scrum makes sure that the team members come forward to address the complex problems. Scrum uses in the following ways-

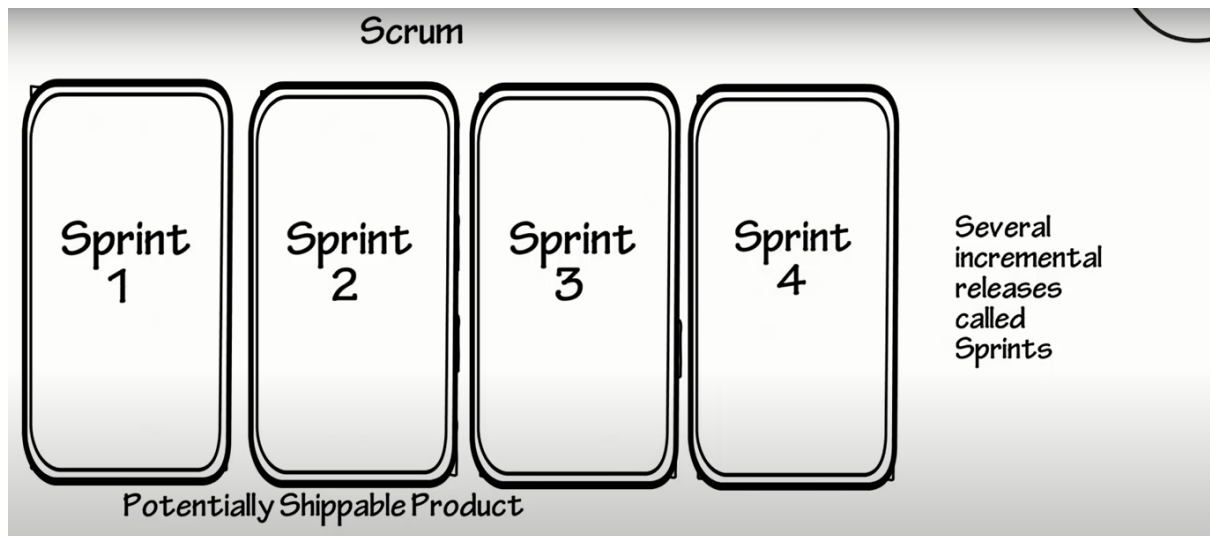
- Software development in Scrum starts with a prioritized list of features- Product Backlog. The team discusses on:
 - The backlog
 - What is not yet completed
 - The duration required to complete the task.
- **Sprint is the heart of Scrum and has the following features-**
 - ✓ The goal of each Sprint is to create a quality product.
 - ✓ Each Sprint ends with a Sprint Review.
 - ✓ Once the first Sprint is over, the team selects the work items to develop a new Sprint.
 - ✓ Sprint is continued till the project deadline is met.
 - ✓ Sprint is a timeboxed activity when the actual product is developed.
 - ✓ The duration of Sprint is from **one week to one month** to finish items from the product backlog.

- In daily Scrum, teams meet to discuss the progress of the work.
 - The Daily Scrum should be **no longer than 15 minutes**.
 - Each team member should be prepared to discuss the status of the work.
 - The **Scrum Master** facilitates the meeting for the team members to achieve the common goal.
- **What are the Benefits Received from the Scrum Methodology?**
 - ✓ Higher productivity
 - ✓ Better-quality products
 - ✓ Reduced time to market
 - ✓ Better team dynamics
 - ✓ Happier employees

Scrum vs Waterfall







Sprint:

The **heart of Scrum** is a Sprint, a time-box of two weeks or one month during which a potentially releasable product increment is created. A new Sprint starts immediately after the conclusion of the previous Sprint. **Sprints consist of the Sprint planning, daily scrums, the development work, the Sprint review, and the Sprint retrospective.**

- ✓ In **Sprint planning**, the work to be performed in the Sprint is planned collaboratively by the Scrum Team.
- ✓ The **Daily Scrum Meeting** is a 15-minute time-boxed event for the Scrum Team to synchronize the activities and create a plan for that day.
- ✓ A **Sprint Review** is held at the end of the Sprint to inspect the Increment and make changes to the Product Backlog, if needed.
- ✓ The **Sprint Retrospective** occurs after the Sprint Review and prior to the next Sprint Planning. In this meeting, the Scrum Team is to inspect itself and create a plan for improvements to be enacted during the subsequent Sprint.

Scrum Theory:

Scrum was built on **empirical process** control theory which is also known as **Empiricism**. Empiricism says that knowledge is gained from the experience in creating the decisions based on what we know. Scrum operates on how to control the risk and **optimize** predictability. These three pillars support each and every implementation of empirical process control: **adaptation, transparency, and inspection.**

Transparency:

We have many important aspects for all Scrum processes that need to be viewable to those who are responsible for the deliverables. This requires all parts to be defined by common standards. So, the viewers share a common understanding of what is being observed.

For example:

- ✓ A common language is used to refer to the process that must be shared by all participants; and
- ✓ Those who perform the work and those who accept the result of increment must need to share a common definition of "Done".

Inspection:

Those who work on Scrum must need to regularly inspect the **Scrum artifacts** and progress towards the Sprint Goal to detect **undesirable variations in the timeline**. This inspection should be done more often so that it does not **disrupt** the productive work time. From the point of work, inspection ensures the profits when they are carefully performed by highly skilled inspectors.

Adaptation:

By following the above steps, inspection reveals that one or more parts inside the process deviate the outside acceptance criteria, and so, that the delivered product will not be acceptable, so the team needs to do adjustments to the process or the material being processed. The sooner the adjustments, the better minimization from further deviation.

Scrum Values:

Originally, the Scrum guide was based on the three pillars of Scrum. Those three pillars are- **Transparency, Inspection, and Adaptation**. These pillars were ensuring the team to work collaboratively, adapt to the changes and respect each other's decisions during work.

Scrum isn't a strategy. It is a non-repeatable kind of process. Scrum is a framework of standards, roles, and practices. The framework enables individuals and organizations to find out what works best for them. The Scrum processes are specific and stick to the time and context.

Along with the processes of Scrum, there come the core Scrum Values upon which the Scrum framework is based. These are:

- ✓ Commitment,
- ✓ Focus,
- ✓ Openness,
- ✓ Respect, and
- ✓ Courage

Here is a detailed view of Scrum values, how they guide our work, actions, and behaviour with respect to Scrum.

1. Commitment:

Scrum teams must be committed to progress and willing to have practical objectives and stick to them. This is a team activity where you are a part of a team, and you are accountable to work together and to conform to your commitments.

This value can be expressed in three ways-

- ✓ **Sprint-based commitment-** Sprint reflects the realistic goals and a short time duration to achieve these goals. Thus, the team has to be committed to their tasks in order to achieve the Sprint goals.
- ✓ **Commitment as a team-** As a team, you need to welcome the changes and represent the adaptability. As a team, you can achieve the project goals in smaller chunks. In case of any issue raised, the team can gather and discuss with each other to come up with a concrete solution to tackle the issue.
- ✓ **Commitment as an individual-** Being an individual in a team, you should contribute as much as you can to achieve the Sprint goal. This represents your commitment as an individual. As an individual, you can exhibit a commitment to learning, collaboration, following Scrum values, working software, and so on.

2. Focus:

An iterative-incremental approach and timely delivery in Scrum helps to keep us stay focussed towards the project goal. Once the requirements are clear and the goal is set, the most effective way to attain the goal is to be goal-oriented. This motivates you for delivering faster, better and yield more. By focussing more on a goal, you can avoid resource wastage and deliver on time. This Scrum value leverages lower rate of risks and provides ample time to improve and deliver the needful.

3. Openness:

The Scrum **induction** requires transparency and openness. We need to investigate reality with a specific end goal to make sensible adjustments. Team members should be open about their work, progress, what they learned and the issues they are facing. Also, you should be open to working with colleagues, recognizing individuals to be individuals, and not assets, robots or replaceable pieces of hardware.

You should always be open to team up with the stakeholders in terms of orders and abilities. You should be open to work together with team members in any extensive condition. As a team member, you should be open towards sharing feedback and gaining from each other.

4. Respect:

As a part of the Scrum team, you should respect colleagues, their decisions, and their experience. As an efficient team member, you should also respect diversity. You should respect your stakeholders by not building anything in which people are not interested. You need to equally respect your users by resolving their problems. As a responsible team member, you need to completely adhere to the Scrum framework and the associated Scrum roles. As a part of the Scrum team, you should respect each other's skills, knowledge, and insights.

5. Courage:

Adaptability to change forms the bedrock of any Scrum project and to accept a change, courage is needed.

Scrum is all about taking risks and finding out an optimized solution. The Scrum team is allowed to think of different approaches to workshop the best and most appropriate solutions. In order to implement new things to the project, we need to explain these new ideas to the team.

What is Scrum Team?

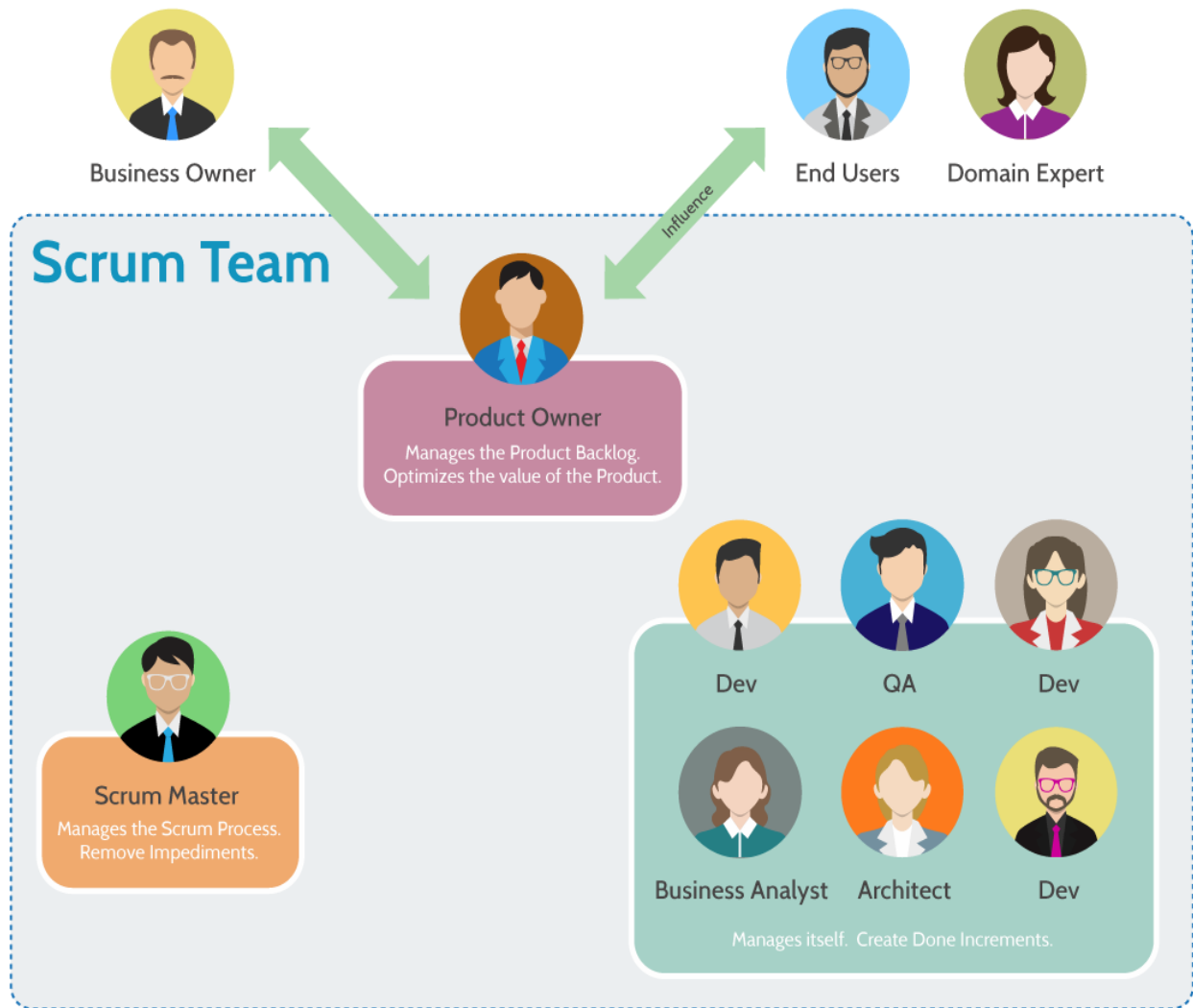
A **Scrum** Team is a collection of individuals (typically between five and nine members) working together to deliver the required product increments. The Scrum framework encourages a high level of communication among team members, so that the team can:

- ✓ Follow a common goal
- ✓ adhere the same norms and rules
- ✓ show respect to each other

Structure of Scrum Team

The Scrum Team. The Scrum team includes the:

- ✓ Product Owner
- ✓ Scrum Master, and
- ✓ Development team



What is Scrum Team:

The Scrum Team share different tasks and responsibilities related to the delivery of the product. Each role are closely inter-related. It is recommended for Scrum team members work together in the same location whenever possible. Let's take a look at each of these roles in terms of their responsibilities, authority, and characteristics.

Product Owner:

The Product Owner is the Team member who knows what the customer wants and the relative business value of those wants. He or she can then translate the customer's wants and values back to the Scrum team. The Product Owner must know the business case for the product and what features the customers' wants. He must be available to consult with the team to make sure they are correctly implementing the product vision. Most importantly, he must have the authority to make all decisions necessary to complete the project, in other words, the Product Owner is responsible for managing the Product Backlog which includes:

- ✓ Expressing Product Backlog items clearly.
- ✓ Ordering the Product Backlog items to best achieve goals and missions.
- ✓ Optimizing the value of the work the Team performs.
- ✓ Ensuring that the Product Backlog is visible, transparent, and clear to all, and shows what the Team will work on further.
- ✓ Ensuring that the Team understands items in the Product Backlog to the level needed.

Scrum Master:

The scrum master help to keep the team accountable to their commitments to the business and also remove any roadblocks that might impede the team's productivity. They met with the team on a regular basis to review work and deliverables, most often in a weekly cadence. The role of a scrum master is to coach and motivate team member, not enforce rules to them. The role of a scrum master includes:

- ✓ ensure the process run smoothly
- ✓ remove obstacles that impact productivity
- ✓ organize the critical events and meeting

Development Team:

Development Teams are structured and empowered by the organization to organize and manage their own work. The resulting synergy optimizes the Development Team's overall efficiency and effectiveness. Development Teams have the following characteristics:

- ✓ They are self-organizing. No one (not even the Scrum Master) tells the Development Team how to turn Product Backlog into Increments of potentially releasable functionality
- ✓ Development Teams are cross-functional, with all the skills as a team necessary to create a product Increment
- ✓ Scrum recognizes no titles for Development Team members, regardless of the work being performed by the person
- ✓ Scrum recognizes no sub-teams in the Development Team, regardless of domains that need to be addressed like testing, architecture, operations or business analysis
- ✓ Individual Development Team members may have specialized skills and areas of focus, but accountability belongs to the Development Team as a whole.

Scrum events

The Scrum events are key elements of the Scrum Framework. They provide regular opportunities for enacting the Scrum pillars of Inspection, Adaptation and Transparency. In addition, they help teams keep aligned with the Sprint and Product Goals, improve Developer productivity, remove impediments and reduce the need to schedule too many additional meetings.

There are five Scrum events (the **Sprint**, **Sprint Planning**, **Daily Scrum**, **Sprint Review**, **Sprint Retrospective**), each with their own purpose, time constraints and participants.

Purpose

While learning the details of each is very important for effective Scrum, at the highest level, the purpose of each is actually quite simple:

- ✓ **Sprint** - All work in Scrum is done in a series of short projects called Sprints. This enables rapid feedback loops.
- ✓ **Sprint Planning** - The Sprint starts with a planning session in which the Developers plan the work they intend to do in the Sprint. This plan creates a shared understanding and alignment among the team.
- ✓ **Daily Scrum** - The Developers meet daily to inspect their progress toward the Sprint Goal, discuss any challenges they've run into and tweak their plan for the next day as needed.
- ✓ **Sprint Review** - At the end of the Sprint, the Sprint Team meets with stakeholders to show what they have accomplished and get feedback.
- ✓ **Sprint Retrospective** - Finally, the Scrum Team gets together to discuss how the Sprint went and if there are things they could do differently and improve in the next Sprint.

Time constraints

To help create discipline and focus, each of the Scrum events has a predefined time constraint, or timebox:

- ✓ The Sprint's timebox is no greater than a month, though they are typically two weeks in duration.
- ✓ The **Sprint Planning**, **Sprint Review** and **Sprint Retrospective** are timeboxed to 8 hours, 4 hours and 3 hours, respectively, for a one month Sprint. The team generally sets the maximum duration of these events to be less when the Sprint is shorter than a month.

- ✓ The Daily Scrum is 15 minutes regardless of the length of the Sprint.

These timeboxes enable productive meetings that encourage relevant discussions while conversations not related to the event's purpose are discouraged. If the team is able to achieve the purpose of the time-boxed events (Sprint Planning, Daily Scrum, Sprint Review and Sprint Retrospective) **before the maximum time allotted is reached**, they should simply end the meeting.

When teams are unable to achieve the purpose of the event in the timebox, the team should investigate where they can find opportunities to improve how they conduct their events.

Driving focus into the timeboxed events is a discipline that allows team members to spend less time in meetings, freeing them to spend more time doing other work.

Scrum Events Quick Reference

The following is a quick summary of the timeboxed Scrum events. The Sprint is a container for all of these events and is required to have a maximum duration of one month or less.

<u>Event</u>	<u>Inspection</u>	<u>Adaption</u>	<u>Participants</u>	<u>Timebox</u>
Sprint Planning	Product Backlog, Product Goal, Definition of Done	Sprint Backlog, Sprint Goal	Scrum Team	8 hours for a 1 month Sprint
Daily Scrum	Progress toward Sprint Goal	Sprint Backlog	Developers	15 minutes
Sprint Review	Increment, Sprint, Product Backlog, Progress toward Product Goal	Product Backlog	Scrum Team, Stakeholders	4 hours for a 1 month Sprint
Sprint Retrospective	Sprint, Definition of Done	Actionable improvements, Definition of Done	Scrum Team	3 hours for a 1 month Sprint

Scrum Artifacts:→

What Are the 3 Artifacts of Scrum?

Scrum is a popular Agile project management methodology that can help your team effectively complete projects, especially when quick **turnarounds** and multiple deliveries are required. One of the foundations of the Scrum framework is its **three artifacts**. These three artifacts of Scrum help capture and share vital project information with project teams.

A little background on Scrum

Scrum is an Agile project management framework that **embraces** an iterative approach to project management.

Teams new to Scrum may find it difficult to adopt because the framework has many unique roles, elements, and terms. For instance, a Scrum master, rather than a project manager, leads Scrum teams. The Scrum master's main job is to support the team's autonomy and remove obstacles the team encounters.

Despite its unique aspects, Scrum shares many similarities with other Agile frameworks. Projects are broken down into sprints that typically last one to four weeks, and each sprint ends with the delivery of a workable version of the final project deliverable. Scrum Masters host daily standups throughout each sprint to keep the team focused and on the same page.

What are Scrum artifacts?

Scrum artifacts are simply tools that help teams implement the Scrum framework and complete projects successfully.

Scrum artifacts help ensure your team accomplishes what it needs to within each project sprint and for the final project deliverable. The three artifacts of Scrum enable you to identify and document the work that must be done for the project to be successful.

What are the 3 Artifacts of Scrum:→

The three Scrum artifacts are:

- ✓ Product backlog
- ✓ Sprint backlog
- ✓ Product increment

Let's now look at these artifacts in detail.

1. Product backlog

The product backlog is an ordered list of the features and requirements needed to complete your project. The list can include bugs or defects that need to be resolved, features that must be added, and other technical or project work your team has to complete before the project is over. The order of the list is based on

priority — with the items at the top being either the most critical tasks or tasks requiring immediate completion.

The product backlog helps ensure everyone knows what the project requirements are and which ones have priority. It also helps your team address the remaining project scope requirements and how much time may be needed to complete them.

Features are generally listed in the backlog as user stories, which describe how the features will be useful to the customer.

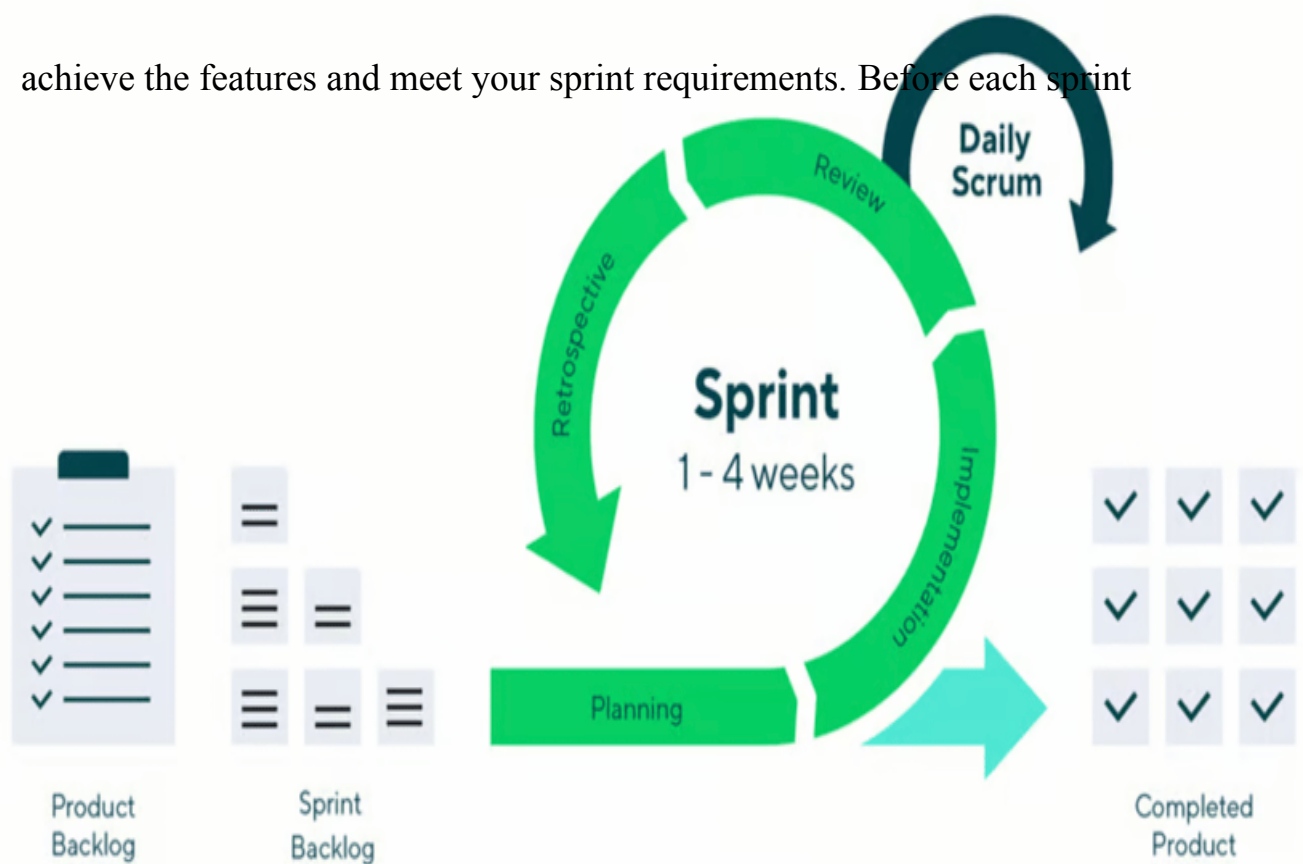
The product backlog is often called a “**living artifact**” since it is continuously updated to reflect changes in the project and customer feedback. The process of updating the product backlog is called **product backlog refinement** or **backlog grooming**.

2. Sprint backlog

Where the product backlog lists everything that needs to be accomplished before the end of the project, the sprint backlog includes only what needs to be completed during the current sprint.

Since the sprint backlog covers your most current activities, it often has greater detail and more complete user stories than the rest of the product backlog. Your sprint backlog should include your release plan explaining how you intend to

achieve the features and meet your sprint requirements. Before each sprint



begins, your team will conduct a sprint planning meeting where you review the

product backlog with the product owner and determine which requirements will move into the sprint backlog and which will have to wait for future sprints.

As with the product backlog, the sprint backlog is a living document that is regularly reviewed and updated as changes occur. Items can be removed if they become unnecessary and new items added as requirements change.

3. Product increment

The product increment is often thought of as the most critical of the three artifacts in Scrum. It's the version of the product that will be delivered at the end of each sprint.

While the **sprint backlog outlines what must be completed within a sprint**, the **product increment details the outcome of all that work**. The increment is a **prototype**, draft, or working version of the final product expected by the customer.

At the end of the initial sprint, your team will deliver product version one, including all of that sprint's goals. At the end of the second sprint, version two will be delivered. Version two of the product increment should include all of the features and requirements already delivered in version one and any changes, new features, and requirements added in the second sprint.

Each product increment should include the features of the current sprint and fully integrate all backlog items completed to date, from all past sprints.

Artifact transparency:

Scrum **relies** on transparency. **Decisions to optimize value and control risk are made based on the perceived state of the artifacts.** To the extent that transparency is complete, these decisions have a sound basis. To the extent that the artifacts are incompletely transparent, these decisions can be **flawed**, value may diminish and risk may increase.

The **Scrum Master** must work with the Product Owner, Development Team, and other involved parties to understand if the artifacts are completely transparent. There are practices for coping with incomplete transparency; the Scrum Master must help everyone apply the most appropriate practices in the absence of complete transparency. A Scrum Master can detect incomplete transparency by inspecting the artifacts, **sensing patterns**, listening closely to what is being said, and detecting differences between expected and real results.

The Scrum Master's job is to work with the Scrum Team and the organization to increase the transparency of the artifacts. This work usually involves

learning, convincing, and change. Transparency doesn't occur overnight, but is a path.

Why Artifact Transparency Matters

1. **Enhanced Collaboration:** Artifact transparency fosters collaboration among the Scrum Team members and stakeholders. When everyone has access to the same information, it becomes easier to align efforts and make informed decisions.
2. **Informed Decision-Making:** Transparency empowers stakeholders and product owners to make informed decisions about the product's direction and priorities. This is particularly important as it ensures that the product stays aligned with customer needs and market dynamics.
3. **Early Detection of Issues:** With transparent artifacts, issues and obstacles become visible early in the process. This allows the Scrum Team to address them promptly, reducing the risk of delays and surprises.
4. **Continuous Improvement:** Scrum is all about inspecting and adapting. Transparent artifacts provide the data needed to conduct effective Sprint Reviews and Sprint Retrospectives, which, in turn, drive continuous improvement within the team.
5. **Building Trust:** Transparency builds trust among team members and stakeholders. When everyone can see what is happening and why, trust is **cultivated**, leading to smoother collaboration and more successful projects.

Creating Artifact Transparency:

Achieving artifact transparency requires a commitment to the Scrum values of **openness, inspection, and adaptation**.

Here are some practical steps to ensure transparency in Scrum artifacts:

- **Regular Updates:** Keep Scrum artifacts up-to-date in real-time. The **Product Backlog, Sprint Backlog, and Increment** should reflect the current state of the project.
- **Clear Documentation:** Maintain clear and concise documentation for each artifact. Use tools like digital boards or project management software to ensure easy access for all stakeholders.

- **Frequent Inspections:** Encourage frequent inspections of artifacts during Scrum events, especially during the Sprint Review and Sprint Retrospective. This allows for feedback and improvements.
- **Open Communication:** Promote open and honest communication within the Scrum Team and with stakeholders. Encourage questions, discussions, and clarifications regarding artifacts.